

MAHMOUD ALKWISEM

SOFTWARE ENGINEER | MACHINE LEARNING ENGINEER

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PROFESSIONAL EXPERIENCE

Machine Learning Developer | Fani's Lab

Sep 2024 - Present

- Co-developed LADy, a Python toolkit for latent aspect detection in online reviews using NLP techniques.
- Integrated LDA, CTM and neural models for topic modeling; helped design an Arabic review dataset for model training.

Software Development Team Lead | Vosyn

Jul - Oct 2023

- Led a 12+ person team delivering a voice synthesis platform with Python, PyTorch and Tortoise TTS.
- Integrated Llama 2 and Transformers; presented engineering progress weekly to 5+ stakeholders.
- Introduced Jira task tracking and workflows, contributing to a reported 20% productivity increase.

SELECTED PROJECTS

Production-Grade RAG Pipeline

LlamaIndex | Qdrant | Ragas

- Built retrieval over 10,000+ pages of financial and medical PDFs; hybrid BM25/dense search plus Cohere reranking improved precision 34%.
- Automated evaluation reached 92% average faithfulness and under 3% hallucinations; parent-child chunking reduced token use 25%.

Multi-Agent Autonomous Pipeline

LangGraph | Python | Pydantic

- Orchestrated four research agents with guarded state transitions, asynchronous execution and human review, reducing manual research time 85%.
- Enforced structured outputs with Pydantic, achieving a reported 99.8% successful execution rate across API model updates.

End-to-End MLOps Pipeline

XGBoost | DVC | MLflow

- Built a fraud detection pipeline for streaming transactions; DVC versioned data across 50+ training iterations.
- Automated Docker/GitHub Actions releases to under 12 minutes; MLflow tracking and drift checks triggered retraining.

TECHNICAL SKILLS

Languages: Python, Go, Java, C, JavaScript, SQL | **AI/ML:** PyTorch, XGBoost, Transformers, LlamaIndex, LangGraph, Ragas, MLflow

Backend & data: Node.js, Spring Boot, Qdrant, PostgreSQL, MySQL, MongoDB | **Tools:** Docker, DVC, GitHub Actions, Git, GCP, Azure

AI development tools: Claude Code, OpenAI Codex

EDUCATION

University of Windsor | Bachelor of Software Engineering, Computer Science

2021 - 2025

Little Scholarship (2021) | Coursework: Data Structures & Algorithms, NLP/Transformers, Databases